Model R: formulation

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What the model does

At each decision moment, Model R assigns every free ambulance to a base. It uses four goals in a strict order of priority, so a lower goal is only used to choose among the plans that are best for all higher goals:

1. leave as few areas as possible that no ambulance can reach within $2T$ (the $2T$ rule);
2. then, cause as few minutes of crew overtime as possible;
3. then, make as few moves above the limit M as possible;
4. then, maximise expected coverage within T, minus the cost of moving.

Defaults: $T = 30$ min, so $2T = 60$ min; $q = 0.3$; $\alpha = 0.97$; $C_j = 2$.

1. Sets

Symbol	Meaning
I, index i	Free ambulances at the decision moment
J, index j	Bases
V, index v	Areas where emergencies happen
$S \subseteq J$	Bases with a sleeping crew; no free ambulance may be sent there

2. Parameters (given)

Symbol	Meaning	Unit
d_v	Share of emergencies in area v; $\sum_v d_v = 1$	fraction
t_{jv}	Driving time from base j to area v	min
$g_{jj'}$	Driving time from base j to base j'; $g_{jj} = 0$	min
$\delta_i \in J$	Base where ambulance i is now, or is already heading to	–
$h_i \in J$	Home base of ambulance i	–
s_i	Time left in ambulance i's shift	min
e_j	Ambulances at base j that cannot be moved (sleeping crews); they still answer calls	count
C_j	Capacity of base j	count
M	Number of moves allowed without penalty	count
T	Response-time limit; the outer limit is $2T$	min
q	Probability that an ambulance is busy, $0 < q < 1$	–
α	Required probability of reaching an area within $2T$	–
cv	Coefficient of variation of travel times (an assumption)	–
$\lambda \geq 0$	Penalty per minute of moving	per min
$P \geq 0$	Penalty per area with no ambulance within T ($P = 0$ switches it off)	per area

3. Derived parameters (computed before solving)

Travel times are assumed normally distributed, with mean t_{jv} and standard deviation $cv \cdot t_{jv}$. Then the chance constraint $\Pr(\text{travel time}_{jv} \leq 2T) \geq \alpha$ is equivalent to $t_{jv} + z_\alpha \, cv \, t_{jv} \leq 2T$, where z_α is the standard normal quantile ($z_{0.95} = 1.645$, $z_{0.97} = 1.881$, $z_{0.99} = 2.326$).

$$a_{jv} = \begin{cases} 1 & \text{if } t_{jv} \leq T \\ 0 & \text{otherwise} \end{cases} \qquad \text{base } j \text{ reaches area } v \text{ within } T$$

$$b_{jv} = \begin{cases} 1 & \text{if } t_{jv}\,(1 + z_\alpha \, cv) \leq 2T \\ 0 & \text{otherwise} \end{cases} \qquad \text{base } j \text{ reaches } v \text{ within } 2T \text{ with prob. } \geq \alpha$$

$$V_2 = \left\{ v \in V : \textstyle\sum_{j \in J} b_{jv} \geq 1 \right\} \qquad \text{areas that at least one base can reach within } 2T$$

$$c_{ij} = g_{\delta_i j} \qquad \text{moving time of ambulance } i \text{ to base } j \text{ (0 if it stays)}$$

$$r_{ij} = c_{ij} + g_{j h_i} \qquad \text{time to reach base } j \text{ and then drive home}$$

$$ot_{ij} = \lceil \max(0,\ r_{ij} - s_i) \rceil \qquad \text{overtime minutes if ambulance } i \text{ goes to base } j$$

$$\bar{k}_v = \min\left(|I| + \textstyle\sum_{j \in J} e_j,\ \ \sum_{j \in J} a_{jv} \, C_j \right) \qquad \text{most ambulances that can ever cover area } v$$

c_{ij} uses the ambulance's destination base, not its exact position on the road (the "teleportation" assumption of van Buuren et al., 2018). Areas outside V_2 cannot meet the $2T$ rule from any base, so the rule is not applied to them.

4. Decision variables

Variable	Domain	Meaning
x_{ij}	$\{0,1\},\quad i \in I,\ j \in J$	1 if ambulance i is assigned to base j
y_{vk}	$\{0,1\},\quad v \in V,\ k = 1, \dots, \bar{k}_v$	1 if area v is covered within T by at least k ambulances
w_v	$\{0,1\},\quad v \in V_2$	1 if area v breaks the $2T$ rule
u_v	$\{0,1\},\quad v \in V$	1 if area v has no ambulance within T (only needed if $P > 0$)
o	integer, $o \geq 0$	number of moves above M

Ambulance i stays where it is when $x_{i\delta_i} = 1$; any other choice is a move.

To keep the formulas short, we write

$$n_j \;=\; \sum_{i \in I} x_{ij} + e_j \qquad \text{for every } j \in J$$

for the total number of ambulances at base j after the decision (free ones placed there plus the fixed ones).

5. Objective

The four goals are optimised lexicographically (in strict order):

$$\operatorname{lex\,min}\ (\, f_1,\ f_2,\ f_3,\ -f_4\,)$$

where

$$f_1 = \sum_{v \in V_2} w_v \qquad \text{areas breaking the } 2T \text{ rule (minimise)}$$

$$f_2 = \sum_{i \in I} \sum_{j \in J} ot_{ij}\, x_{ij} \qquad \text{overtime minutes (minimise)}$$

$$f_3 = o \qquad \text{moves above } M \text{ (minimise)}$$

$$f_4 = \sum_{v \in V} \sum_{k=1}^{\bar{k}_v} d_v\,(1-q)\,q^{k-1}\,y_{vk} \qquad \text{(maximise)}$$
$$\quad -\ \lambda \sum_{i \in I} \sum_{j \in J} c_{ij}\, x_{ij} \;-\; P \sum_{v \in V} u_v$$

f_4 is the expected coverage within T, minus the cost of moving and the penalty for areas with no ambulance within T.

How to read f_4. Its first term is the expected coverage of MEXCLP (Daskin, 1983). If n ambulances can reach area v within T, the chance that at least one of them is free is $1 - q^n$. The k-th of those ambulances adds $(1-q)\,q^{k-1}$ to that chance, and these terms add up to $1 - q^n$. Because the terms shrink as k grows, the solver always switches on $y_{v1}, y_{v2}, \dots$ in order, so no extra ordering constraint is needed.

6. Constraints

$$\sum_{j \in J} x_{ij} = 1 \qquad \forall\, i \in I \tag{1}$$

$$\sum_{k=1}^{\bar{k}_v} y_{vk} \;\le\; \sum_{j \in J} a_{jv}\, n_j \qquad \forall\, v \in V \tag{2}$$

$$n_j \le C_j \qquad \forall\, j \in J \tag{3}$$

$$\sum_{i \in I} (1 - x_{i\delta_i}) \;\le\; M + o \tag{4$'$}$$

$$x_{ij} = 0 \qquad \forall\, i \in I,\; j \in S \tag{6}$$

$$u_v \;\ge\; 1 - \sum_{j \in J} a_{jv}\, n_j \qquad \forall\, v \in V \quad \text{(only if } P > 0) \tag{7}$$

$$\sum_{j \in J} b_{jv}\, n_j \;\ge\; 1 - w_v \qquad \forall\, v \in V_2 \tag{9E}$$

$$x_{ij},\, y_{vk},\, w_v,\, u_v \in \{0, 1\}, \qquad o \in \mathbb{Z}_{\ge 0}$$

In words.

- **(1)** Every free ambulance goes to exactly one base.
- **(2)** Area v can count at most as many covering ambulances as there are ambulances at bases within T of it.
- **(3)** No base holds more ambulances than its capacity.
- **(4′)** At most M ambulances may move for free; every move above M increases o, which is penalised in f_3. An ambulance whose current base is a sleeping base has $x_{i\delta_i} = 0$ by (6), so it always counts as a move.
- **(6)** No free ambulance is sent to a base where a crew is sleeping.
- **(7)** If no ambulance is within T of area v, then $u_v = 1$ and the penalty P applies. The right-hand side is at most 0 as soon as one ambulance is in range, so u_v is then free to be 0.
- **(9E)** Every area in V_2 should have at least one ambulance at a base that reaches it within $2T$ with probability α. If that is impossible, $w_v = 1$ and the area counts in f_1.

Numbering. Labels follow the earlier models so they can be compared. Compared with the strict Model E, the move limit (4) is softened into (4′), and the shift rule (5), "$x_{ij} = 0$ if $j \neq h_i$ and $r_{ij} > s_i$", is removed: lateness is now measured by the overtime goal f_2 instead of being forbidden.

7. Feasibility

Model R has a solution if and only if every base can hold its fixed ambulances and the bases without a sleeping crew have room for all free ambulances:

$$e_j \le C_j \quad \forall\, j \in J \qquad \text{and} \qquad \sum_{j \in J \setminus S} (C_j - e_j) \;\ge\; |I|.$$

Why. Constraints (1), (3) and (6) are the only ones that can conflict, and they ask exactly this: put each free ambulance on a free spot at a base without a sleeping crew. Every other constraint can always be met by setting $y = 0$, $w = 1$, $u = 1$ and $o = |I|$.

8. Solution method

The model is solved with Gurobi's hierarchical multi-objective (`setObjectiveN` with priorities, relative tolerance 0, MIP gap 0). One weighted sum of the four goals is avoided: the penalties would need weights around 10^4 against coverage coefficients around 10^{-5}, so with Gurobi's relative MIP gap the coverage part would only be solved approximately. On the Fez-Meknes instances (18–40 bases, 199 areas, up to 33 ambulances), each solve takes under 0.2 s.